

Sprint 1 Deliverable

Vision Statement

Executive Summary

The goal of this report is to provide an overarching summary, goals, and who will be affected by our project about a grocery pathfinding website. The goal of this project is to mitigate the time to navigate a grocery store to find your desired items. People have other responsibilities outside of grocery shopping or have to catch a bus, and this website is intended to help them get in and out of a grocery store as quickly as possible so they can continue their other duties. This report will define clear stakeholders, features, and risks that will be important to achieving and driving towards the final product. With these factors determined, a project schedule will be developed, and an update on the project's current status and individual contributions will be provided.

Project Description

Grocery Finder is a website intended to provide users with a smooth and quick experience navigating a grocery store. The user will be able to choose between optimizing grocery costs, the type of grocery store, or the quality of the grocery store. Based on this choice, the website will allow the user to create a shopping list with their desired groceries or grocery store inventory. When the user finishes this list, they can click a button for an efficient path to all items, allowing them to locate items quickly and reduce unnecessary backtracking. The user will also be given the option to show their location, so the website can show nearby grocery stores and provide directions to them depending on how they want to. As the user navigates to the

different item locations, they can check off the items to better keep track of what they have and haven't got yet.

Business Case

Grocery Finder will benefit the stakeholders and the user community by providing a centralized platform that helps users reduce their grocery costs and shopping time. The application will allow users to compare basket prices across multiple stores in their area so they can select the most affordable option before beginning their trip. Once inside the store, users will be guided through an optimized route with labeled aisles and the items needed to be grabbed in each aisle allowing them to quickly locate items while minimizing unnecessary backtracking.

The current market contains applications that support price comparison as well as tools that assist with in-store navigation; however, most solutions focus on only one of these functions rather than combining them into a single experience. For example, price-comparison platforms such as Instacart and Flipp allow users to view weekly deals and compare store pricing but do not provide in-store route optimization. On the other hand, certain retailer-specific apps like Walmart and Target offer aisle locations within their own stores but do not allow cross-store basket cost comparison. Because these solutions are fragmented, users must switch between multiple applications to fully plan and execute a cost-efficient trip. Grocery Finder differentiates itself by combining cross-store price aggregation with pathfinding-based aisle navigation in one unified interface, reducing friction and improving overall usability.

There are, however, product risks that must be considered. The application depends on accurate and frequently updated pricing data, which may vary by location and change rapidly

due to promotions or supply fluctuations. If pricing data is outdated or incomplete, user trust in the platform could decline. Additionally, implementing store-specific aisle maps requires cooperation from retailers or consistent data scraping, both of which present operational and technical challenges. Competitive pressure from established grocery chains expanding their own mobile features also presents a risk. Despite these challenges, the integrated value proposition and focus on both cost and time efficiency provide a strong opportunity for differentiation within the digital grocery market.

As shoppers increasingly rely on technology to improve everyday efficiency, Grocery Finder offers a competitive advantage through its unified approach to trip planning and execution. This positions the project as a practical, user-focused solution capable of enhancing the overall grocery shopping experience while remaining aware of competitive and operational risks within the industry.

Stakeholder Description

Alan Lee (Product Owner)

Alan Lee is the primary stakeholder and Product Owner of this project as it is his idea and his name is at the forefront of the development of this software. Alan wants to make sure that we develop and deliver the best software that makes the user's grocery shopping experience as seamless and efficient as possible.

Sebastian Grammas, Peijing Xu, Qizheng Sui (Will)

As project stakeholders/developers we have a stake in the completion of this project and delivering an app that benefits the users making grocery shopping more efficient and less time-

consuming. We also have a commitment to the product owner to complete his vision of the application.

Professor Sturman

Professor Sturman has a stake in this project as our professor and educator in helping guide us to building the best software possible through lecture materials and advice.

Namir Xi (weekly meeting TA)

Namir has a stake in this project as our weekly meeting TA through helping guide us through our weekly status reports. They will help us stay on track and give us advice on what the next best steps are.

Major Features

Feature	Feature Description	User Benefit
Nearest Grocery Store	The app will identify nearby grocery stores based on the zip code entered by the user.	This provides users with a simple and efficient way to locate convenient shopping options while also supporting price comparisons across stores so they can determine where items are available at

		the lowest cost.
Grocery Cart/Inventory Shopping	The app will provide users with access to grocery store inventory, allowing them to build their shopping cart before arriving at the store.	This allows users to prepare their shopping list in advance, reducing in-store decision making and improving overall trip efficiency. Users can also compare their cart across multiple stores to estimate total costs and identify the most cost-effective shopping option.
Efficient Grocery Store Pathfinding	The app will generate an optimized route through the grocery store based on the items in the user's shopping cart and their locations within the store. Using aisle data and store layout information, the app will guide users step-by-	This helps users complete their shopping trip more quickly by reducing time spent searching for items and preventing inefficient movement throughout the store. As a result, users experience a more organized

	step to each item while minimizing unnecessary backtracking.	and less stressful shopping process.
Coupon/Deals Available	The app will display available coupons, promotions, and store deals for items in the user's shopping cart, such as multi-item discounts (e.g., 2 for \$5) or limited-time price reductions. These offers will be integrated into the shopping experience so users can easily identify savings opportunities while building their cart or selecting a store.	This helps users lower their overall grocery expenses by making discounts more visible and easier to apply during the planning process. By understanding which stores offer the best promotions, users can make more informed purchasing decisions and maximize their savings.

Major Risks

1. Data Acquisition & API Connection

Risk: Grocery stores that we plan on including on our website (Aldi, Hannaford, Aldi, Walmart) may not have public APIs, or their data may be behind paywalls or scrapers.

- Consequence of Non-Mitigation: The application becomes a shell with no real-time pricing or inventory, meaning the core function of finding groceries becomes useless.
- Mitigation Strategy: Conduct research to see available APIs and see if we could obtain API test keys or web-scraping libraries. If not possible, see if we could use a third-party aggregator API.

2. Algorithmic Complexity (Pathfinding)

Risk: Implementing an inefficient pathfinding algorithm that takes too long or incorrectly finds the optimal path

- Consequence of Non-Mitigation: The app may provide optimal paths that are actually longer, leading to high computational latency on the front end.
- Mitigation Strategy: Research established algorithms like A* or Dijkstra's. Start with a simpler algorithm to see the performance, and if there is high latency, use more complex algorithms to try and reduce latency

3. Technical Debt & Framework Learning Curve

Risk: Team members have varying levels of familiarity with web frameworks

- Consequence of Non-Mitigation: Slower development speeds, code that is hard to debug, and missed milestones, leading to an unfinished product.
- Mitigation Strategy: Do research on the web frameworks (React and Flask) and discuss with the group what was used to learn the frameworks to see if some methods are more useful and could be more beneficial for the group.

4. Project Management & Workflow

Risk: Ineffective use of Jira for task tracking and ticket management.

- Consequence of Non-Mitigation: Overlapping work, forgotten tasks, and an inability to track project health, leading to potential cramming of the project.
- Mitigation Strategy: Appoint someone to check in with each groupmate about their current progress on their assigned tasks and/or have weekly group meetings to discuss what needs to be done, what needs attention, or if someone needs help with a task .

5. Deployment & Hardware Bottlenecks

Risk: Developing for mobile without a developer account or struggling with hosting environments for a web app.

- Consequence of Non-Mitigation: There may be a working codebase, but there's a chance that nobody can actually see or use it, leading to not being able to present it to the stakeholders
- Mitigation Strategy: For mobile, we could use Expo Go for React to bypass Apple's developer mode requirement. In terms of hosting environments, we could talk to the professor about how we could host our app.

User Scenarios



Josh Smith is a 25-year-old graduate student studying theatre arts while completing his degree. He holds a bachelor's degree and currently relies on graduate funding along with savings from summer work to support himself. In his free time, Josh enjoys playing video games, eating, drinking, and partying with friends. His primary motivations are to successfully graduate from his program and manage the demands of student life day to day.

Josh is looking to get in and out of the grocery store as fast as possible because he needs to get home and study. He wants to use Grocery Pathfinder so that he can walk down the grocery aisles and grab what he needs most efficiently to conserve time. Josh begins by opening the website and logging into his account through the login button, then in the store selection page that signing in takes him to, he selects that he is going to Hannafords taking him to the next page, where he then searches for each item he is buying and puts it into his cart in the app while taking the bus to the store. After he completes his cart by selecting every item he wants, Josh will hit the cart complete button to signal that he is done updating his shopping cart. Then the website will take him to a page where it displays each item in the cart, telling him where to go in the store, updating in real time what direction he needs to walk upon entering to have the most efficient route through the store. After he has put an item into his cart in the store, he checks off that he has grabbed that item, and the app updates the path to the next item. This way, he does

not end up walking around aimlessly and/or going down aisles multiple times looking for a specific item.



Susan Sugar is a 66-year-old retired nurse who lives in a small home with her 30 cats. She completed nursing school and now relies on Social Security as her primary source of income. Susan enjoys cooking, baking, and knitting in her free time, and she finds purpose in caring for her cats and knitting sweaters for her grandchildren. Her main motivations are to live out her remaining years peacefully while ensuring the well-being of her cats and staying connected to her family through the things she loves.

Susan Sugar has run out of cat food for her 30 cats and is in desperate need of finding more, but she doesn't want to leave her cats alone for too long, so she decided to enter and leave the store as quickly as possible. Susan frequently gets confused about which aisle she needs to go to and gets sidetracked easily, causing her to forget her grocery list. Susan Sugar will begin using the product by logging into the grocery website. Then she can enter the grocery store name and the item she wants into the website page after the login, afterwards she can create a shopping cart and add items that she desires into the cart, once she completes her shopping list she can hit the complete button to take her to the next page, since she wants to enter and leave as quickly as possible she can click on the fastest route button to optimize her shopping cart to get the most

optimal route to pick up on all of her items, then she will be able to get the exact location of the item as well as the most optimal path. The app will update the path in real time as she approaches the destination of the item. Once the item is obtained, Susan checks off the item on the app/website, and the app/website will redirect her to the next item on her list.

Project Schedule

Sprint # :	Date:	Plan:	Deliverable :	Interim Features to Be Developed :	Est. team hour
Sprint 1 :	2/2-2/13	Finalize scope, split tasks, begin Vision statement, add user scenarios, start API feasibility research.	Sprint 1 docs draft + API research in progress	Initial feature list + priority ranking Early store/API feasibility notes (Sam's Club / alternatives) Draft user flow for grocery list → route output	35-50 hrs
Sprint 2 :	2/10 - 2/21	Finish Week 1 deliverables, confirm store decision criteria and fallback if aisle data is blocked.	Week 1 deliverable package and UI rough draft aligned to scenarios	Store selection criteria defined Fallback strategy defined (if aisle data unavailable) UI rough draft aligned with user scenarios	35-55 hrs
Sprint 3 :	2/21 - 2/28	Define data model, set repo structure, implement list front-end or API, wire basic navigation flow.	Running app skeleton and list mock data	Repo/project scaffold created - Basic shopping list input UI/API endpoint - Mock data flow (input → list display)	45-70 hrs

				- Initial data model schema	
Sprint 4 :	2/28 - 3/10	Implement “sort by aisle” using a mock aisle map for a pilot store, add store selection and unknown-item handling; draft connector interface for stores.	End to end MVP (mock version)	Mock aisle map for one pilot store “Sort by aisle” logic working on mock data Store selection UI/control Unknown-item handling behavior Draft Storeconnectorinterface	55-80 hrs
Sprint 5 :	3/10 - 3/16	Integrate pilot store data, build item, cache results, basic logging.	Live data prototype	Pilot store data ingestion module Product/item cache (basic caching layer) Lookup result persistence Basic request/error logging	60-90 hrs
Sprint 6 :	3/17 - 3/23	Harden integration, error handling, retries/timeouts, tests unit and simple e2e, basic CI checks; UI states (loading/empty/error/etc.)	Integration and tests/CI	Retry/timeout handling for external lookups Loading/empty/error UI states Unit tests + simple end-to-end flow test Basic CI pipeline checks	55-85 hrs
Sprint 7 :	3/24 - 3/30	add 2nd store connector or improve mapping accuracy and fallback rules; refactor to clean “StoreConnector” architecture.	Multi-store foundation	Multi-store connector foundation or improved mapping accuracy Refactored connector abstraction Improved fallback rules (category/unknown items)	55-85 hrs
Sprint 8 :	3/31 - 4/6	MVP demo readiness: full user flow polished, scenario-based demo script, performance cleanup, data consistency	MVP demo (ready to present)	Full end-to-end user flow polished Demo-ready scenario(s) Data consistency checks across stores	45-70 hrs

		across stores.		Basic performance cleanup (latency/caching tuning)	
Sprint 9 :	4/7 - 4/13	UX polish, better grouping, “not found” suggestions, small quality- of-life features.	Polished UI and QoL features	Better item grouping in list/route view “Item not found” suggestion flow QoL improvements (edit/reorder/clear behaviors as applicable)	45 -75 hrs
Sprint 10 :	4/14 - 4/20	bug bash, final testing, deployment packaging, presentation outline.	v1.0 release candidate and docs	Release candidate stabilization Deployment/package scripts/docs Final regression test checklist Presentation-ready screenshots/demo path	45-70 hrs
Sprint 11 :	4/21 - 4/27	v1.1 + buffer: remaining fixes, optional extra store, final report/slides, final demo rehearsal and submission packaging.	v1.1 and final submission package	Final bug fixes from testing/demo feedback Optional extra store support Final demo rehearsal polish Submission packaging validation	35 - 60 hrs

Project Status Report

The project has made decent progress for the past three weeks as the group has created a solid foundation that awaits further development. The group has made rigorous progress towards what the grocery finder application should look like, what its functions are supposed to be, as

well as creating a framework, a skeletal structure, for how the project will look and function and will be expanded upon in the coming months. The project team has first created a solid foundation of what our project, the Grocery Finder application, has. These include finding the closest grocery store near the user's location, the ability to customize the user interface, the ability to create a grocery list, optimization based on cost and price for a grocery list, as well as information regarding the stores and markets that may appear on our application, which includes phone numbers and websites. The team began by identifying the target customers of our product, which for our product, as a tool to optimize groceries, is applicable to basically everybody. The team then created multiple user scenarios which includes a customer of our product, their reason for using our product, and what specific feature of our product that they require. Take Josh Smith, for example, the twenty five years old grad student who is majoring in theatre and living out of an apartment who wants to spend less time buying groceries would want to use our product to spend less time shopping for groceries and more time on his course work and curriculum. This process helped the team to summarize and complete the product's major requirements through visualizing how a customer would potentially use our product. From the user scenarios, the team generated user stories based off of specific use cases that potential customers from our target audience could utilize to fulfill their needs. Finally, with a structured set of user requirements, we ranked them based on their importance to the overall product and finalized them into our major measureables, which includes having a system that enables user to create an account, login and log off of the account whenever they want to, the product must also display the closest grocery store based on zip code and optimize a grocery list based on cost, travel time between stores, and both.

Contribution Summary

Peijing Xu

I worked on finalizing the project status report section as well as the user scenarios section of Sprint 1 and contributed the most during the generation of our product's use cases, coming up with potential customers who would use our product in various different situations. I also participated in all in class assignments and was always on time to the weekly one on one meetings

Sebastian Grammas

I have helped with all the in class assignments. I contributed the most during the Use Cases lecture using Figma. I wrote and planned out the business case, doing research on the market landscape of other apps out there currently. I also wrote out the stakeholder descriptions for the Vision statements and built out the major features and fine tuned them to be more descriptive.

Qizheng Sui:

I created the team's project schedule including sprints, end dates, feature mapping, and time estimates, took charge on Sam's Club API feasibility research, and led our most recent team meeting by organizing updates and assigning next steps. I also helped the team with in-class assignments to keep our work aligned with course requirements.

Alan Lee

I helped with all the assignments assigned in class, set up the Jira board, set up the group's GitHub, and organized meetings outside of class. I am the group's product owner, so I've been helping facilitate group discussion and ensuring that the group is on track with developing towards the final product. I also took part in researching the Target API and how the

group could potentially implement it for the project. I wrote the executive summary and project description of the vision statement.